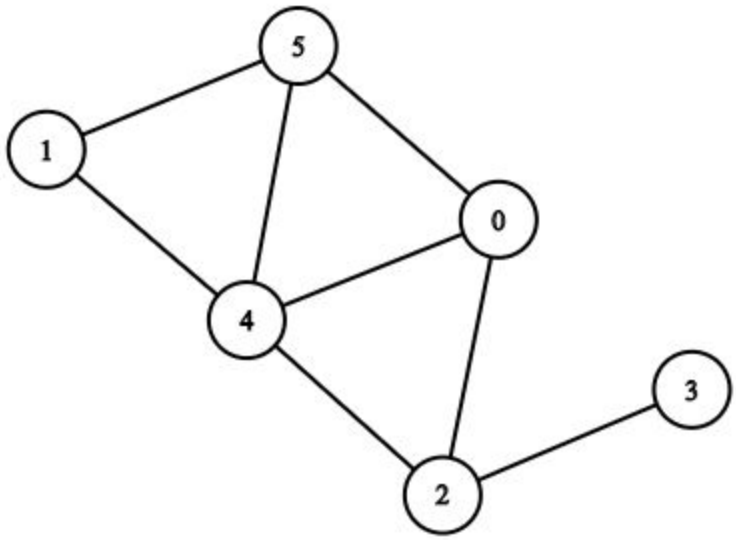
1. Suppose you have the following constraint graph and you need to find an assignment for each of the variables so that constraint satisfaction problem can be solved. A wise man comes along to help you and offers to tell you the value of any one of the variables. Which of the following variables will you choose so that your search procedure simplifies the most?



Ans. 4

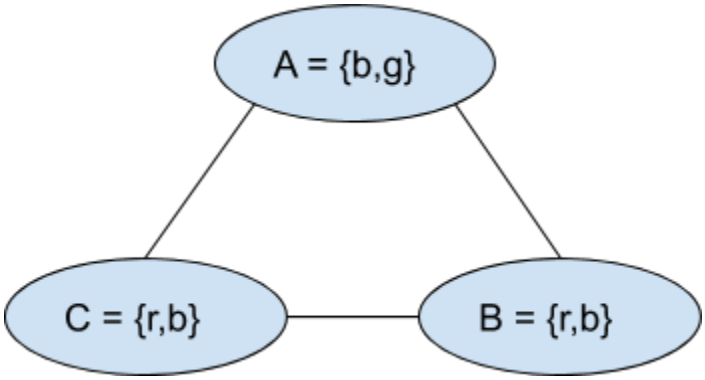
If we know the value of node 4 then the CSP becomes a linear chain and we know that acyclic CSP (tree-like CSP) can be solved in polynomial time. Thus, knowing the value of node 4 will simplify the CSP the most.

2. Which of the following consistency checks can propagate information between two unassigned variables?
    a. Forward consistency
    b. Arc consistency
    c. None of the above

In arc consistency, we check that for a certain value of first node is there any legal value for second node. Arc consistency can be used to prune possible values for unassigned nodes.

3. After performing arc consistency we can solve CSP without requiring any backtracking. State whether this is True or False
    a. True
    b. False

    Consider the following graph where edges represent diff constraint. This graph is arc consistent as for each value of a node its neighbour has some value which they can take.



    However, we know that A can only take value g since only one of the B and C can take value r while other will have to take b.
    If we try to assign b to A, then we will assign r to B but will be left with no value to assign for C and would be forced to backtrack. Therefore arc consistency does not ensure that we will not have to backtrack.

4. What is the worst-case time complexity for solving a CSP with n variables where each variable can take d values?
    a. $O(nd^2)$
    b. $O(n^d)$
    c. $O(d^n)$
    d. $O(n^2d)$

    We would need to do backtracking search which is dfs procedure with branching factor d and depth n thus time complexity will be $O(d^n)$

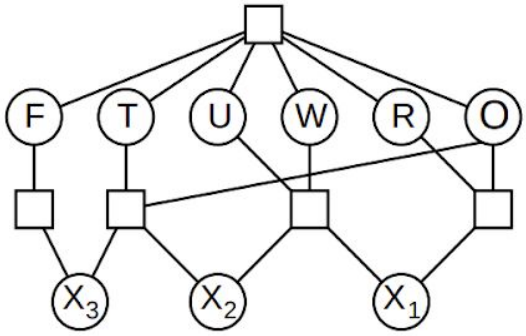
5. You are given a CSP with n variables and each variable can take d values. You are able to decompose this problem into two independent subproblems each having n/2 variables. What is the worst-case time complexity of solving this CSP using the information of decomposition?
    a. $O(d^{n/2})$
    b. O(nd)

c. $O(d^n)$
d. $O((n/2)^d)$

Since we are able to decompose the problem into two independent subproblems now we can solve the two subproblems and then compose their solutions. So our algorithm will involve solving CSP of n/2 nodes twice - thus **$O(d^{n/2})$**

6. Consider the following cryptarithmetic problem where TWO and FOUR are three and four digit numbers respectively. Which of the following constraint(s) is/are correct for modeling this problem? Assume that X_1, X_2 & X_3 are carry overs resulting from the additions at the unit, tens and the hundreds place. Also, all digits are distinct.

```
  T W O
+ T W O
-------
F O U R
```



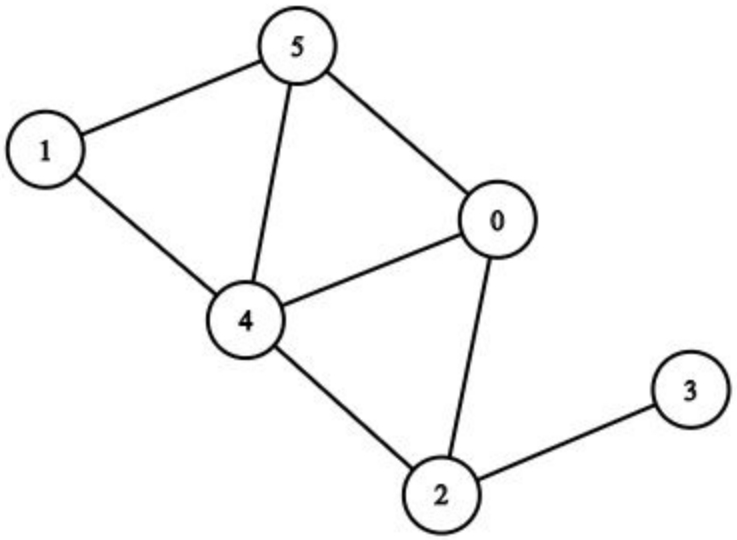
a. $X_1 + W + W = U + 10.X_2$
b. $X_2 + T + T = O + 10.F$
c. T > 0, F > 0
d. alldiff(F, T, U, W, R, O, X_1, X_2, X_3)

We have the following constraints for the problem:

$O + O = R + 10.X_1$	using the sum at units place
$X_1 + W + W = U + 10.X_2$	using the sum at tens place
$X_2 + T + T = O + 10.X_3$	using the sum at hundreds place
$X_3 = F$	using the sum at thousands place
$T > 0$	since TWO is a 3 digit number
$F > 0$	since FOUR is a 4 digit number
alldiff(F, T, U, W, R, O)	since the digits are distinct

Therefore, (a), (b), (c) are correct.

7. Suppose you have the following constraint graph and you need to find an assignment for each of the variables so that constraint satisfaction problem can be solved. Each node can take values from the set {‘a’, ‘b’, ‘c’} and nodes connected by an edge need to have a different value. Using the minimum remaining value heuristic, find the order of assignments of nodes. (For example. If 1 is assigned a value of a, then 3 is assigned a value of c then report your answer as 1a3c). For node tie breaking, a smaller numbered node is prefered over a higher numbered node. For value assignment, ‘a’ is assigned before ‘b’ and ‘c’, and ‘b’ is assigned before ‘c’.

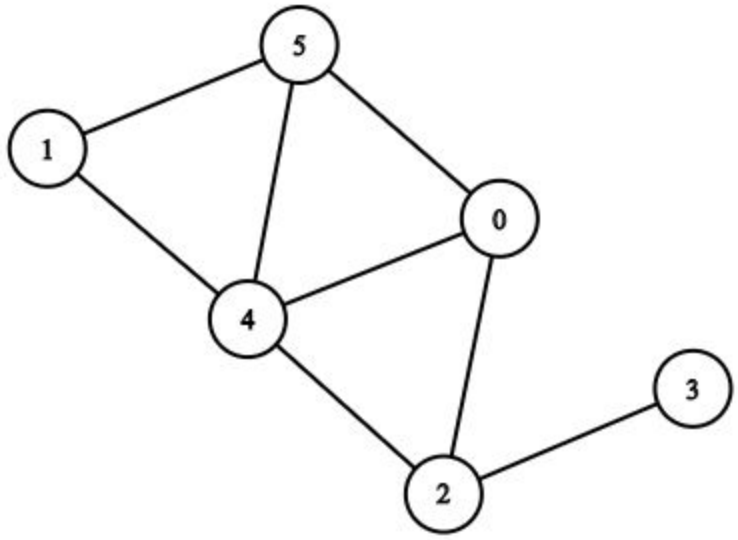


Ans: 0a2b4c5b1a3a

The following table shows the remaining values of all nodes at every step.

Step	0	1	2	3	4	5	Assignment
1	3 (abc)	3 (abc)	3 (abc)	3 (abc)	3 (abc)	3 (abc)	0 - a
2	-	3 (abc)	2 (bc)	3 (abc)	2 (bc)	2 (bc)	2 - b
3	-	3 (abc)	-	2 (ac)	1 (c)	2 (bc)	4 - c
4	-	2 (ab)	-	2 (ac)	-	1 (b)	5 - b
5	-	1 (a)	-	1 (a)	-	-	1 - a
6	-	-	-	1 (a)	-	-	3 - a

8. Suppose you have the following constraint graph and you need to find an assignment for each of the variables so that constraint satisfaction problem can be solved. Each node can take values from the set {‘a’, ‘b’, ‘c’} and nodes connected by an edge need to have a different value. Using the least constraining value heuristic, find the order of assignments of nodes. (For example. If 1 is assigned a value of a, then 3 is assigned a value of c then report your answer as 1a3c). For node ordering, a smaller numbered node is prefered over a higher numbered node. For value tie breaking, ‘a’ is assigned before ‘b’ and ‘c’, and ‘b’ is assigned before ‘c’.



Ans: **0a1a2b3a4c5b**

The following table shows the constraints caused by each value assignment at every step. Node column is lexicographically ordered. ‘-’ means, that value assignment leads to failures.

Step	Node	a	b	c	Assignment
1	0	3	3	3	0 - a
2	1	1	2	2	1 - a
3	2	-	2	2	2 - b
4	3	0	-	0	3 - a
5	4	-	-	1	4 - c
6	5	-	0	-	5 - b

9. Consider the following constraint satisfaction problem. Which of the following statements is true?



   a. The arc is consistent in both directions.
   b. The arc is not consistent in both directions.
   c. The arc {x, y} is consistent but arc {y, x} is not consistent.
   d. The arc {x, y} is not consistent but arc {y, x} is consistent.

   The arc from {x, y} is not consistent because for x = 1 there exists no value of y such that 1 > y. The arc from {y, x} is not consistent because for y = 5, 6 there exists no value of x such that x > 5 or x > 6.

10. Which of the following statement(s) is/are true for CSPs?
   a. Tree structured CSPs cannot be solved in polynomial time.
   b. Forward checking is the same as arc consistency.
   c. Computation time depends on no. of constraints and variables.
   d. Iterative min-conflicts algorithm can solve 10000-queens.

   Answers to each of these options are given in the slides. Please refer to the following pages:
   (a) : 34
   (b) : 22 - 30
   (c) : 39
   (d) : 39